

Experiment 9: Nmap Timing & Performance (IDS Evasion)

Aim

To understand Nmap timing templates used for performance control and IDS evasion:

- -T0 (Paranoid) / -T1 (Sneaky) / -T2 (Polite) – Slower scans, less likely to trigger IDS
- -T3 (Normal) – Default timing
- -T4 (Aggressive) / -T5 (Insane) – Faster scans

Procedure

1. Open Terminal.
2. Perform a polite scan (slower, IDS-friendly):
`nmap -T2 -p 80 scanme.nmap.org`
3. Perform a normal scan (default timing):
`nmap -T3 -p 80 scanme.nmap.org`
4. Optional – Compare with aggressive timing:
`nmap -T4 -p 80 scanme.nmap.org`
5. Observe the time taken by each scan (shown at the end of output).

Sample Output / Results

(i) Polite Scan (-T2):

```
$ nmap -T2 -p 80 scanme.nmap.org
Starting Nmap 7.94 ...
PORT      STATE SERVICE
80/tcp    open  http
Nmap done: 1 IP address (1 host up) scanned in 1.25 seconds
```

(ii) Normal Scan (-T3):

```
$ nmap -T3 -p 80 scanme.nmap.org
PORT      STATE SERVICE
80/tcp    open  http
Nmap done: 1 IP address (1 host up) scanned in 0.35 seconds
```

Observation: Polite (-T2) is slower and generates less traffic, making it less likely to trigger Intrusion Detection Systems. Normal (-T3) is the default balanced mode.

```

kali@kali: ~
Session Actions Edit View Help
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 10:36 -0400
Stats: 0:00:21 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 0.35% done
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
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Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.46s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 988 closed tcp ports (reset)
PORT      STATE      SERVICE
22/tcp    open      ssh
80/tcp    open      http
256/tcp   filtered  fw1-secureremote
445/tcp   filtered  microsoft-ds
554/tcp   filtered  rtsp
1025/tcp  filtered  NFS-or-IIS
1310/tcp  filtered  husky
2099/tcp  filtered  h2250-annex-g
2967/tcp  filtered  symantec-av
5060/tcp  open      sip
5061/tcp  open      sip-tls
5900/tcp  filtered  vnc

Nmap done: 1 IP address (1 host up) scanned in 59.98 seconds

(kali@kali)-[~]
$ nmap -T3 scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 10:38 -0400
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.014s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 988 filtered tcp ports (no-response)
PORT      STATE      SERVICE
80/tcp    open      http
83/tcp    closed    mit-ml-dev
139/tcp   closed    netbios-ssn
143/tcp   closed    imap
161/tcp   closed    snmp
443/tcp   closed    https
3389/tcp  closed    ms-wbt-server
5002/tcp  closed    rfe
5060/tcp  open      sip
5061/tcp  open      sip-tls
8080/tcp  closed    http-proxy
8089/tcp  closed    unknown

Nmap done: 1 IP address (1 host up) scanned in 41.65 seconds

```